Educational Phlogiston

Why a Formula Is Never Just a Formula

A reflection on cognitive load, symbolic representation and the teaching of physics equations

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Cognitive load is a genuine constraint on learning. Yet when it is invoked whenever students fail to interpret a formula, it can become an invisible explanatory substance: educational phlogiston. The difficulty may lie less in the number of symbols than in the network of physical meanings those symbols conceal.

THE phrase *cognitive load* has acquired an unusual explanatory power in education. When students fail to follow an explanation, the material imposed too much cognitive load. When a diagram contains too many labels, it imposed extraneous cognitive load. When a formula sheet supplies both words and symbols, it may create split attention or redundancy.

The concept is often useful. It can also become a kind of **educational phlogiston**: an invisible substance invoked whenever learning fails, without identifying what the learner actually failed to perceive, represent or coordinate.

Consider the apparently elementary equation $F = ma$, whose symbolic economy is deceptive consisting of three letters and one equality sign, yet almost none of its physical content is visibly present. Indeed, every attempt to make the equation more explicit makes it symbolically heavier:

$$F = ma, \qquad F_{\text{net}} = ma, \qquad \sum F = ma,$$
$$\sum_i \mathbf{F}_i = m\mathbf{a}, \qquad \sum_i \mathbf{F}_i = \frac{d\mathbf{p}}{dt}.$$

Each successive form is, in one sense, a better representation. It tells us that the relevant force is the net force, that several interactions may contribute, that force and acceleration are vectors, and finally that ma is a restricted constant-mass form of the more general momentum law.

Yet each clarification also introduces another demand: a subscript, a summation sign, vector notation, an index, a derivative or a new physical quantity.

"Every clarification improves the physics while increasing the symbolic burden."

Cognitive Load Theory would describe part of this as increasing intrinsic complexity. The student must coordinate more interacting elements at once. Split-attention research further suggests that learning is impaired when mutually dependent sources must be searched and mentally integrated across separated locations [2].

A formula sheet that places a verbal equation beside its algebraic form may reduce that search by spatially integrating the representations.

	energy transferred = charge flow × potential difference	$E = Q\,V$
HT	**potential difference across primary coil × current in primary coil = potential difference across secondary coil × current in secondary coil**	$\boldsymbol{V_p\,I_p = V_s\,I_s}$
	density = mass / volume	$\rho = \frac{m}{V}$
	thermal energy for a change of state = mass × specific latent heat	$E = m\,L$
	weight = mass × gravitational field strength	$W = m\,g$
	work done = force × distance (along the line of action of the force)	$W = F\,s$
	force = spring constant × extension	$F = k\,e$
	distance travelled = speed × time	$s = v\,t$
	acceleration = change in velocity / time taken	$a = \frac{\Delta v}{t}$
	(final velocity)^2 – (initial velocity)^2 = 2 × acceleration × distance	$v^2 - u^2 = 2\,a\,s$
	resultant force = mass × acceleration	$F = m\,a$
HT	**momentum = mass × velocity**	$\boldsymbol{p = m\,v}$
	period = 1 / frequency	$T = \frac{1}{f}$
	wave speed = frequency × wavelength	$v = f\,\lambda$
HT	**force on a conductor (at right angles to a magnetic field) carrying a current = magnetic flux density × current × length**	$\boldsymbol{F = B\,I\,l}$

Figure 1: Extract from AQA GCSE Physics eqn. sheet.

Nevertheless, integration on the page does not guarantee integration in the mind.

The student still has to understand that the m in $F = ma$ is not mean merely "mass" as a word but rather denoting (and connoting) the numerical measure of a particular system's inertial mass, expressed in a chosen unit.

Similarly, a does not merely mean "acceleration". It may represent a changing speed, a changing direction, or both. In circular motion at constant speed, the student must accept that an object is accelerating even though the magnitude of its velocity remains unchanged.

The equation therefore requires more than symbol recognition. It requires the student to bind a symbol to a quantity, a quantity to a physical interpretation, and that interpretation to a particular system and coordinate frame.

The reversal error

This is close to the mechanism exposed by the classic reversal-error research. Clement asked students to translate statements such as "there are six times as many students as professors" into algebra. Many wrote

$$6S = P$$

rather than

$$S = 6P.$$

The difficulty persisted even among mathematically experienced, science-oriented students. Clement identified two common processes: syntactic matching, in which symbols are arranged according to word order, and a static-comparison interpretation, in which the coefficient is attached to the larger group rather than used as a multiplier relating two numerical quantities [1]. The letters were being treated partly as labels for nouns rather than as variables denoting measured numbers.

The same ambiguity is easy to overlook in physics. A learner may read $F = ma$ as a compressed sentence:

> *Force is mass times acceleration.*

An expert reads something much richer:

> Relative to an inertial frame, the vector sum of the external forces acting on the selected system equals the time rate of change of its momentum; for constant inertial mass this becomes $m\mathbf{a}$.

These are different modes of reading, with the first treating the equation as a verbal substitution template. The second treats it as a structured representation of a physical model.

Sherin describes expert understanding of physics equations in terms of *symbolic forms*: recurring symbolic structures associated with conceptual schemas [5]. A physicist may see ma as an amount of inertia coupled to a rate of change of motion, or see

$$\sum_i \mathbf{F}_i$$

as the accumulation of distinct interactions into a resultant.

The expert does not decode every symbol independently. A conceptual structure and a symbolic pattern are activated together. This chunking reduces the experienced load, although the external notation has not changed.

What the left-hand side hides

The equation's left-hand side is especially underrepresented. The single symbol F, even when replaced by F_{net}, is a *placeholder* for physically different interactions:

$$\mathbf{F}_{\text{net}} = \mathbf{F}_{\text{gravity}} + \mathbf{F}_{\text{normal}} + \mathbf{F}_{\text{friction}} + \mathbf{F}_{\text{tension}} + \mathbf{F}_{\text{electric}} + \mathbf{F}_{\text{magnetic}} + \cdots$$

These forces do not arise from a common visible mechanism merely because they occupy the same algebraic position. Some depend on position, some on velocity, some on deformation, some on contact constraints and some on charge.

The summation sign compresses this physical heterogeneity into a single mathematical operation.

The right-hand side is similarly compressed. The acceleration might be linear acceleration along a track, gravitational free fall, tangential acceleration from a changing speed, or centripetal acceleration from a changing direction.

In a uniform magnetic field, for example,

$$q\mathbf{v} \times \mathbf{B} = m\mathbf{a}$$

may produce circular motion, with

$$qvB = \frac{mv^2}{r}.$$

Here the same a must be interpreted geometrically as v^2/r, while the force is interpreted dynamically as the magnetic part of the Lorentz force. The charge q determines the interaction, while the inertial mass m determines the response.

In uniform gravitational free fall,

$$m_g \mathbf{g} = m_i \mathbf{a},$$

and the observed independence of free-fall acceleration from the body's mass depends upon the empirical equivalence of gravitational and inertial mass:

$$m_g = m_i.$$

Thus the humble m in $F = ma$ silently carries conceptual history. It is not simply the "quantity of matter" suggested by everyday language.

A network rather than a dictionary

There is also an element of circularity, although it needs stating carefully. Force, mass and acceleration are not defined independently by dictionary-style descriptions and then connected by experiment.

Acceleration can be defined kinematically from changes in velocity, but inertial mass is operationally established through dynamical comparisons, while force is characterised through the changes of motion associated with interactions.

In elementary presentations, force is often defined through $F = ma$, while mass may subsequently be inferred from

$$m = \frac{F}{a}.$$

The concepts form an interconnected theoretical system. Their meaning arises partly from their relations to one another.

This is not necessarily vicious circularity. Scientific concepts are often defined within networks rather than by reduction to previously understood words. Yet it undermines the expectation that a one-line verbal gloss can make the equation transparent.

"Force equals mass times acceleration" resembles a definition, while functioning simultaneously as a law, an operational relation, a modelling prescription and, in restricted circumstances, a calculational rule.

Dual coding and duplicated language

Dual Coding Theory is sometimes invoked to justify presenting the verbal and algebraic forms together. Paivio's theory, however, principally distinguishes verbal systems from nonverbal or imagistic systems [3, 4].

An algebraic equation is not straightforwardly a picture. Word form and symbolic form may therefore constitute two linguistic or propositional registers rather than two genuinely independent codes. Pairing

$$\text{force} = \text{mass} \times \text{acceleration}$$

with

$$F = ma$$

may aid symbol identification, but it does not supply the missing physical model in duplicating the relation without representing the phenomenon.

A free-body diagram potentially adds something more valuable because it *encodes* spatial direction, system boundary and interaction structure. A motion diagram adds the temporal pattern of velocity and acceleration. Units add another representational layer:

$$[F] = \mathrm{kg\,m\,s^{-2}}, \qquad [m] = \mathrm{kg}, \qquad [a] = \mathrm{m\,s^{-2}}.$$

Dimensions and units complicate the expression, yet they also tether it to measurement. They distinguish the physics equation from the relatively disembodied algebra of x and y.

The dialect of physics

School algebra commonly treats letters as placeholders for unknown numbers within a formally closed problem. Physics requires letters to perform several roles at once.

A symbol may be an unknown, a measurable variable, a constant, a parameter, a vector, a field, a component, an initial value or a rate of change. The same letter may change role across problems. Its value carries dimensions. Its sign depends on a coordinate convention. Its relevance depends on the selected system.

Redish consequently describes mathematics in physics as a distinctive disciplinary dialect in which mathematical form is continually blended with physical knowledge [6]. Kuo and colleagues similarly show how successful physics problem solving blends conceptual reasoning with formal mathematical manipulation rather than alternating cleanly between the two [7].

Students who have learned equations primarily as procedures for finding x are therefore being asked to enter a much denser semiotic practice. That helps explain why formula sheets can produce surprisingly little transfer. The sheet may eliminate recall while leaving interpretation untouched. It can tell the student that

$$F = ma,$$

but it cannot determine:

- what the system is;
- which interactions count as external forces;
- which direction has been chosen as positive;
- whether the acceleration is linear, tangential or centripetal;
- whether the mass can be treated as constant;
- whether the equation should be resolved into components;
- whether a force is absent or merely balanced;
- whether the resulting value is physically plausible.

None of these is a memory of the formula. They are acts of modelling.

"The formula sheet may remove recall while leaving interpretation unresolved."

The danger, then, is that cognitive load becomes a diagnosis where it should be a question. Certainly, $F = ma$ imposes less notational demand than

$$\sum_i \mathbf{F}_i = \frac{d\mathbf{p}}{dt}.$$

Yet the simpler notation can (just) transfer load (offload) rather than remove it. What is omitted from the page must be supplied through interpretation.

The expert experiences the equation as compressed because the suppressed structure is already available. The novice experiences it as sparse because that structure has not yet been built. Torigoe's work on *unpacking* symbolic equations makes this distinction explicit: successful interpretation requires students to recover the relationships and physical commitments condensed into the notation [8].

This is why limited symbolic representations of physical phenomena often fail to travel intact into students' reasoning. Their very elegance depends upon a body of tacit qualifications.

Physics equations are more than miniature descriptions of nature, being rather a highly compressed set of instructions for constructing a model of nature. The student must learn when to unpack them, what to add and what the symbols are permitted to hide.

Cognitive load is real. But to say that a student cannot use $F = ma$ because the task imposed too much cognitive load explains very little by itself. The more productive diagnosis asks which conceptual bindings have not formed:

symbol to quantity,
quantity to unit,
vector to geometry,
interaction to force,
motion to acceleration,
equation to model.

Without that analysis, cognitive load risks becoming educational phlogiston: an invisible explanatory substance inserted wherever meaning failed to ignite.

References

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Physics equations are compressed models. Fluency consists partly in knowing what has been compressed.